

Problem Statement

Online Food Delivery – Order Pricing System

An online food delivery platform wants to automate its billing system. Since customers can place different types of orders, the final bill amount varies depending on the order type.

All orders share common details such as:

- Order ID
- Customer Name
- Base Price

The final price calculation differs depending on whether the order is:

- Dine-In
- Take Away
- Delivery

You are required to implement this system using **Inheritance and Method Overriding**.

Step 1: Create a Base Class FoodOrder

The base class should contain the following attributes:

- orderId (int)
- customerName (String)
- basePrice (double)

The class should include:

- A method **calculateFinalPrice()** that returns the base price.
- A method **displayOrderDetails()** that prints the order details in the required format.

Step 2: Create Subclasses

Extend the FoodOrder class into the following subclasses. Each subclass must override the **calculateFinalPrice()** method.

1. DineInOrder

$\text{finalPrice} = \text{basePrice} + (\text{basePrice} \times 0.10)$

2. TakeAwayOrder

$\text{finalPrice} = \text{basePrice} + 50$

3. DeliveryOrder

$\text{finalPrice} = \text{basePrice} + 40 + (\text{basePrice} \times 0.05)$

Input Format

- 1. choice (Integer) — 1 → DineInOrder, 2 → TakeAwayOrder, 3 → DeliveryOrder
- 2. Order ID (int)
- 3. Customer Name (String)
- 4. Base Price (double)

Constraints

- $1 \leq \text{choice} \leq 3$
- $1 \leq \text{orderId} \leq 10^6$
- $0 \leq \text{basePrice} \leq 10^6$
- Customer name may contain spaces
- Output must be formatted to 2 decimal places
- Invalid input → No output

Output Format

For valid input:

```
Order ID: <orderId>  
Customer Name: <customerName>  
Final Price: <calculatedPrice>
```

For invalid input: No output

Sample Test Cases

Test Case	Sample Input	Sample Output
1	1 101 John Bow 1000	Order ID: 101 Customer Name: John Bow Final Price: 1100.00
2	2 202 Arjun Mehta 500	Order ID: 202 Customer Name: Arjun Mehta Final Price: 550.00
3	3 303 Sneha Sharma 800	Order ID: 303 Customer Name: Sneha Sharma Final Price: 880.00

Hidden Test Cases

Test Case	Input	Expected Output
1	3 -200 Premium Platinum Client 1000000	(No output) orderId < 0
2	1 420 Rahul Verma 999.50	Order ID: 420 Customer Name: Rahul Verma Final Price: 1099.45
3	2 999 Dr. Rahul Verma 499.99	Order ID: 999 Customer Name: Dr. Rahul Verma Final Price: 549.99
4	1 250 Invalid Negative Order -500	(No output) basePrice < 0
5	4 305 Sneha Patel 600	(No output) choice > 3
